

Set A — Keyword-rich
'I was devastated when he left.'
explicit emotion words
90 stimuli · 8 emotions

Set B — Keyword-free
*'The kitchen table sits empty
now.'*
no explicit emotion words
96 vignettes · 8 domains

LLM
6 models tested
Llama-1B · 8B · Gemma-9B
base + instruct variants

Linear Probing
Encoding
AUROC

Activation Patching
Causality
transfer rate

Layer Knockout
Necessity
critical layers

Geometry Analysis
Structure
silhouette · PCA

Two discovered mechanisms

Affect Reception
binary · early layers · universal

Emotion Categorisation
8-class · late layers
scale-sensitive